

ADR-0002: Reconcile conflicting source values; never flat-average

Status: Accepted **Date:** 2026-06-01 **Deciders:** Product owner, Eng lead

Context

The same parameter differs across sources — e.g. a unit's price on the developer site, on MahaRERA, and on MagicBricks are all different, and its area may be quoted as carpet, built-up, or super built-up. The sources also differ in credibility (RERA is authoritative; listing portals are advertised/indicative). The naive instinct is to average the three into one number. That is wrong here: (1) it pollutes a credible value with an unreliable one; (2) the values often measure different things (carpet vs super built-up area; asking vs registered price), so averaging is apples-to-oranges; (3) it hides the disagreement, which is itself the signal (negotiation room / RERA-vs-listing discrepancy).

Decision

Keep **every** source value as an append-only `ParameterObservation` (never overwrite). Reconcile deterministically in three steps:

1. **Normalize** to one canonical basis first (e.g. ₹ per **carpet** sqft using RERA carpet area; tag price definition: asking / quoted / registered).
2. **Consolidate by credibility — never flat-average.** Default method **ANCHOR**: canonical value = the most credible source for that parameter; others shown as variances. Alternatives: **WEIGHTED** (by tier) when one number is required, **RANGE/MEDIAN** to express uncertainty.
3. **Quantify and expose the spread** (`variance_pct` , agreement band). A wide spread is surfaced as a signal, not smoothed away. The consolidated value stores `inputs_ref` to all observations, and the UI/agent **always cite all source values with their tiers**.

Options Considered

Option A: Flat arithmetic mean → single value

Dimension	Assessment
Complexity	Low
Correctness	Poor — mixes tiers, mixes definitions, hides signal
Auditability	Poor

Pros: Trivial. **Cons:** Destroys the credibility signal; misleading for decisions.

Option B: Normalize → anchor by credibility → show spread (chosen)

Dimension	Assessment
Complexity	Medium
Correctness	High — like-for-like, credibility-aware
Auditability	High — all observations retained and cited

Pros: Honest, defensible, turns disagreement into insight. **Cons:** Requires per-parameter normalization rules and an authoritative-source map.

Trade-off Analysis

More modelling work (units, area basis, anchor rules) in exchange for numbers a CxO can defend in a board room. Given sparse credible data, transparency about disagreement *is* the product's trustworthiness.

Consequences

- **Easier:** auditing; surfacing discrepancies; supporting gap-fill and what-if on the same observation model.
- **Harder:** must maintain normalization rules and an authoritative-source-per-parameter map.
- **Revisit:** the default consolidation method per parameter as we learn which sources to trust.

Action Items

1. ☐ `ParameterObservation` (append-only) + `ConsolidatedParameter` (method, value/range, variance, inputs_ref).
2. ☐ Normalization rules: area basis → carpet; price definition tagging.
3. ☐ Authoritative-source map per parameter (drives ANCHOR).
4. ☐ Encode the v2.2 §6 worked example (3BHK) as a reconciliation test fixture.